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GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I010*

Team Name: *Vishwamithra*

Domain: *Climate Action, Water and Environmental Resilience*

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Problem Statement

Every morning, a village relies on its pond for various needs and employs a solar powered sensor node to monitor water quality parameters, including pH and dissolved oxygen, transmitting data via LoRa. Over time, biofilm formation from algae and organic matter compromises sensor accuracy, leading to unreliable readings and false contamination alerts. This issue undermines community trust and increases maintenance costs due to extensive technician travel for cleaning and recalibration. Current cleaning methods are problematic, mechanical wipers may jam, UV cleaning is energy intensive and chemical coatings could degrade and pollute the pond. The primary challenge is to accurately differentiate between actual water contamination and sensor errors from biofouling while ensuring low-maintenance, energy efficient operation in rural settings.

Problem Understanding

- Primary users: State and regional water authorities, including SWSM Tamil Nadu and TNPCB, along with field researchers.
  Affected communities: Villagers and farmers who depend on ponds and tanks for their daily water needs and irrigation.
  Supporting stakeholders: Gram Panchayats, CPCB, ICAR-CIFRI, and C-DAC.
- Algae, bacteria and organic matter form biofilm on submerged probes within days, gradually affecting sensor accuracy.
  Fixed-timer cleaning may be insufficient during algal blooms and unnecessary during periods of low fouling.
  The monitoring system cannot reliably distinguish biofouling induced errors from genuine changes in water quality.
- False alarms and missed contamination events reduce community trust and may put public health at risk.
  Inaccurate, unverified readings weaken water quality assessments, long term trends and regulatory compliance.
  Frequent field visits, recalibration, cleaning, and premature probe replacement increase maintenance costs.

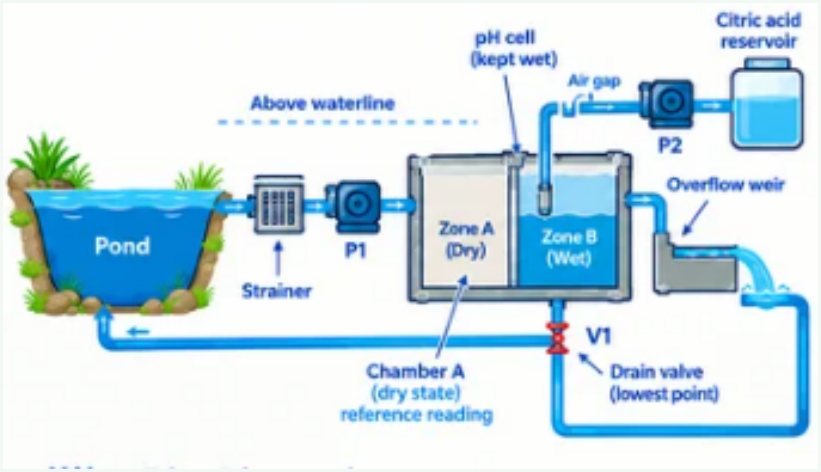
Existing Gaps & Challenges

- Mechanical wipers can jam due to debris and biofilm buildup, UV-C cleaning increases energy consumption during low solar periods, and chemical coatings lose effectiveness over time or risk leaching substances into the pond.
- Conventional monitoring relies on fixed cleaning schedules or raw sensor thresholds, with limited ability to detect sensor drift, distinguish biofouling from genuine contamination or confirm whether cleaning has restored measurement accuracy.
- Developing an affordable, solar-powered node that detects biofouling on device, activates cleaning only when necessary, and verifies sensor recovery using post cleaning measurements without excessive energy consumption or harmful chemical release.

Our Proposed Solution

- Core idea or product: A self diagnosing, solar powered water monitoring node that detects biofouling, distinguishes sensor errors from genuine water quality changes and initiates cleaning only when necessary.
- How the solution works at a high level: Sensors remain dry between measurements where technically feasible, while pH is maintained in a protected wet zone. An STM32L4 calculates a Sensor Health Index using response time, dead time, dry state residuals and 24 hour drift. Based on the detected fouling level, cleaning escalates from an air burst to mechanical wiping, UV-C and citric acid with ultrasound, followed by recovery verification. Sampling intervals adapt to battery status and solar energy availability and readings carry confidence flags.
- Main value delivered to users: More trustworthy water quality data, fewer false alarms, reduced energy consumption, and fewer maintenance visits, targeting six or more months between field visits.

Technical Approach



System Architecture of Vishwamithra

Tech Stack

Frontend: Grafana / ThingsBoard

Backend: LoRaWAN network server and embedded C/C++ firmware on STM32L4

Database: InfluxDB time series database

AI/ML, APIs, or Hardware: Rule based and model fit Sensor Health Index, LoRa radio, solar MPPT, LiFePO battery, and sensor-cleaning actuators

Innovation & Uniqueness

- Dry idle sensor chamber reduces biofouling exposure, while pH remains in a protected wet zone. On-device Sensor Health Index distinguishes suspected sensor fouling from genuine water quality changes. Adaptive cleaning escalates only when needed and verifies recovery through post cleaning measurements.

- Low power, rule based edge processing works without continuous internet connectivity or AI training data. Battery aware sampling and cleaning conserve solar energy and reduce maintenance visits.

- Key Differentiator: Vishwamithra combines on device fouling detection, need based cleaning and post cleaning recovery verification to distinguish unreliable sensor readings from genuine water quality changes delivering more trustworthy data with minimal energy and maintenance.

Feasibility & Scalability

- **Feasibility:** Hardware: STM32L4, water quality sensors, LoRa module, solar panel, MPPT charger, LiFePO battery, and cleaning actuators. Development: Integrate sensing, Sensor Health Index, adaptive cleaning and recovery verification using embedded C/C++. Validation: Conduct bench tests, compare pond readings with manual water samples and perform a six month field trial.
- **Scalability:** Network: Multiple monitoring nodes can communicate through LoRa gateways to a central dashboard. Deployment: Standardised hardware and firmware can be replicated across ponds with site specific sensor calibration. Expansion: The monitoring dashboard can scale from a single village to district wide water monitoring.
- **Risks:** The main risks include low solar energy during monsoon periods, cleaning mechanisms failing to restore sensor accuracy, hardware damage from repeated wet dry cycles and LoRa communication interruptions. These can be mitigated through battery aware sampling and cleaning recovery verification with maintenance alerts when readings remain below 80% of baseline.

Cost for Product (Optional)

- Development or hardware cost: Water quality sensors: 21,500 Self cleaning and fluidics: 10,500 Solar power system: 6,000 STM32L4 controller and PCB: 2,500 IP67 enclosure and 3D printed chamber: 2,500 LoRa module: 800 Total hardware cost per node: 43,800
- Hosting or maintenance cost: Shared LoRa gateway: 8,000 (one time) Server hosting: 6,000/year Consumables and annual maintenance: 3,000/year Recurring annual cost: 9,000
- Estimated total: Approximately 61,000 for the first year, including one monitoring node, one shared gateway, hosting and maintenance. Subsequent annual costs are approximately 9,000, excluding hardware replacement or unexpected repairs.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Gram Panchayats and village water committees monitor pond water quality remotely, receiving clear contamination alerts when water quality changes and maintenance flags when sensors require attention.
- Use case 2: SWSM Tamil Nadu and TNPCB monitor multiple tanks and reservoirs through a central dashboard, prioritising technician visits based on sensor health and failed cleaning recovery rather than fixed maintenance schedules.
- Use case 3: Researchers deploy Vishwamithra at lakes and rivers for long term monitoring, recording sensor drift, cleaning events and confidence flags alongside water quality readings to improve data reliability.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- 6+ months: Target interval between manual cleaning visits. 80%+ recovery: Verify sensor recovery after cleaning; otherwise, trigger a maintenance alert. Less than 1.16 Wh/day: Target energy consumption in optimal power mode. 7–10 days: Target battery autonomy during prolonged cloudy weather.

- More trustworthy water quality alerts, with confidence flags to identify potentially unreliable readings. Fewer unnecessary field visits through condition based maintenance. Traceable readings and cleaning records to support audits and long-term water quality analysis.

- Social: Supports timely identification of potential contamination risks in communities relying on surface water. Economic: Reduces maintenance travel, recalibration expenses, and premature sensor replacement. Environmental: Avoids potentially leaching anti fouling coatings and uses solar powered cleaning methods, including air, light, sound, and mild citric acid.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 6	Target 6.3	Clean Water and Sanitation	Enables reliable water quality monitoring to help identify potential pollution early.
SDG 13	Target 13.1	Climate Action	Uses solar power and adaptive energy management to support resilient water monitoring during challenging weather conditions.

Individual Member Contribution

Member	Contribution or responsibility
Amirthamirra P	Problem research, stakeholder identification, and SDG alignment.
Dheepika Raja	Two zone sensor chamber design and biofouling detection logic.
Harita J	System architecture, STM32L4 firmware logic, and energy-aware power management.
Pooja S	Solar power system design, cost estimation, and dashboard and alert design.

Yukti ID

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